

Luis Mastrangelo

Senior Software Engineer

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Professional Summary

I am Senior Software Engineer with professional and academic experience in Switzerland, Argentina, France and USA. I have worked for more than 15 years in the private industry in different roles, as Software Engineer, Network Administrator and Freelance Developer. I delivered many software projects for a wide variety of clients across different sectors.

I have a Ph.D. in Informatics at the University of Lugano, Switzerland, where I was able to perform outstanding research in programming languages, runtime systems and virtual machines.

In the last years I have been working in the Web3 space. I worked with Ethereum, NEAR, Avalanche, EOS/WAX and Everscale (TON) using mainly Solidity, Rust and TypeScript/JavaScript.

Education

- Apr 2013–June 2019 **Ph.D. Graduate, Computer Science**, *University of Lugano, Switzerland*
Programming languages research in Java, Scala and Haskell, runtime systems and virtual machines with C++, software verification, data analysis and visualization using Python and R.
- Apr 2007–Nov 2011 **Licentiate in Computer Science**, *University of Buenos Aires, Argentina*
This degree is considered by many institutions as being equivalent to an M.Sc. degree.
- Apr 2011–Oct 2011 **Internship, INRIA/University of Strasbourg, France**
Internship title: "A Virtual Machine for Automatic Program Parallelization".
- Apr 2003–Dec 2006 **Computer Analyst**, *University of Buenos Aires, Argentina*
This is a 3 years degree as part of the *Licentiate in Computer Science* programme.

Work Experience

- Nov 2023–Mar 2026 **Senior Software Engineer**, *Hashgraph (ex. Swirlds Labs)*, Remote
Part of the Smart Contracts team, building tools to support EVM development in Hedera networks. While working here I
- created the [Hedera Forking](#) project, which allows developers to fork test their contracts against Hedera System Contracts,
 - contributed to setup a custom Sourcify instance, which allowed developers to verify their Smart Contracts,
 - refactored Hashgraph's JSON-RPC Relay backend so it can be more easily extensible,
 - integrated the Ethereum Execution Client testing framework in Python/Pytest into Hedera's own EVM,
 - debugged production issues while on-call using Grafana and GCP, and
 - added support to onboard new clients to Hashgraph's TheGraph custom instance.

- Apr 2023–May 2023 **Senior Software Engineer**, *ChainSafe*, Remote
I worked in Lodestar, an Ethereum Consensus Client written in TypeScript. I helped to fix some small bugs while working here.
- Jul 2022–Jan 2023 **Senior Blockchain Engineer**, *ConsenSys*, Remote
Part of the Truffle team, a development environment to build and test Solidity Smart Contracts. I focused on their VS Code extension for Truffle. While working here I
 - added and tested new features to the extension using TypeScript, and
 - helped with the release workflow in GitHub Actions.
- May 2022–Jun 2022 **Lead Blockchain Engineer**, *Campground*, Remote
Smart Contract development with Rust and TypeScript for the NEAR blockchain. While working here I
 - built a system in Rust to automatically generate public docs for our smart contracts,
 - tested and deployed the main smart contracts, and
 - helped with the integration of the Smart Contracts into our backend using TypeScript.
- Sep 2020–Apr 2022 **Senior Software Engineer**, *EPAM Systems*, Zurich, Switzerland
Ethereum & NEAR blockchains Smart Contract development with Solidity, Rust and TypeScript. Also architected a greenfield marketplace application.
- Aug 2019–Jan 2020 **Technical Consultant**, *DB&L Services SA for Avaloq*, Lugano, Switzerland
IT infrastructure support for Monitoring Operations at Avaloq Sourcing (CH & LI) SA.
- Dec 2006–Mar 2011 **Project Leader**, *Recursiva S.R.L.*, Buenos Aires, Argentina
Software engineering and architect roles using Java and C# among others technologies. Besides building and testing tasks, I have visited customers and potential customers to gather requirements to implement them in our systems.
- Mar–Jul 2006 **Network Administrator**, *Texpampa and HS Eventos*, Buenos Aires, Argentina
Technical service, software installation and maintenance of company network.
- Mar 2000–May 2004 **Software Developer**, *Fernando Hermo y Asoc.*, Buenos Aires, Argentina
ERP and CRM development using VB 6, SQL Server/Microsoft Access and Crystal Reports. Technical support to customers both on-line and on-site.

Freelance & Open Source Projects

- Apr 2023–Mar 2025 **Software Engineer**, *SEVM Open Source project*, Remote
A Symbolic EVM bytecode interpreter, parser and decompiler written in TypeScript. This work involves parsing and analyzing bytecode to build its corresponding CFG, to then extract method information or decompile to either Solidity or Yul.
- Jan–May 2020 **Software Engineer**, *GTM (cont.) and GTM Dashboard Open Source*, Remote
Added support for submodules to Git Time Metric (Golang) [PR](#) and built a dashboard for GTM from scratch (Rust, TypeScript and Svelte).
- Feb 2014–Jan 2020 **Software Developer**, *JNIF*, Lugano, Switzerland
Java Native Instrumentation Framework written in C++. JNIF allows to analyze and instrument Java bytecode, including JDK objects, something it is not possible with other instrumentation frameworks. It parses the bytecode and reconstruct its CFG so it can be instrumented in later phases.
- May–Sep 2018 **Software Developer**, *Essentia, Family Office*, Lugano, Switzerland
Desktop application written in C# and Scala.
- Apr 2016–Oct 2017 **Software Developer**, *HSC*, Lugano, Switzerland
A Supercompiler (similar to symbolic execution) prototype for a Haskell-like language.

Technical Skills

Programming Languages	Rust, JavaScript/TypeScript, Python, C/C++, Solidity/Yul, Java, Kotlin, Scala, Haskell, Golang, R, C#, Assembler, Prolog, SQL, PHP, HTML/CSS, UML
Paradigms & Fields	OOP, Functional Programming, Static & Dynamic Analysis, Data Science
Databases	PostgreSQL, MySQL, Oracle, Datalog, SQL Server, SQLite, Redis, Design
Editors	VSCode, IntelliJ/CLion/Rider, Eclipse, Visual Studio, Xcode, Vim/Emacs
VCSs	Git, GitHub, GitLab, BitBucket, Mercurial, Subversion, TortoiseSVN, CVS
DevOps	Docker, Podman, Kubernetes, Helm, Lima, GitHub Actions, GitLab CI/CD
Tools	GitHub Copilot, Node.js, LLVM, Apache/Nginx, Azure, GCP, REST, GraphQL
Project Management	Scrum, GitHub Projects, Agile, JIRA, Kanban Boards
Blockchain & Web3	Ethereum, NEAR, Avalanche, Smart Contracts, Foundry, Hardhat, TheGraph

Language Proficiency

English	Proficient User: Level C2 under CEFR
Spanish	Native speaker
Italian	Proficient User: Level C1 under CEFR
French	Independent User: Level B1 under CEFR

Teaching Experience

Apr 2013–Nov 2017	Teaching Assistant , <i>University of Lugano</i> , Switzerland. Multiple courses.
Apr 2006–Nov 2010	Teaching Assistant , <i>University of Buenos Aires</i> , Argentina. Multiple courses.

Publications DBLP

University of Lugano	Luis Mastrangelo , Ph.D. Thesis, Advisors: Prof. M. Hauswirth and Prof. N. Nystrom. When and How Java Developers Give Up Static Type Safety . 2019
University of Lugano	Luis Mastrangelo , M. Hauswirth and N. Nystrom. Casting about in the Dark: A Survey of Cast Usage in Java . 2019
University of Lugano	Luis Mastrangelo , L. Ponzanelli, A. Mocci, M. Lanza, M. Hauswirth and N. Nystrom. Use at Your Own Risk: The Java Unsafe API in the Wild . 2015
University of Lugano	Luis Mastrangelo and M. Hauswirth. JNIF: Java Native Instrumentation Framework . 2014
INRIA/University of Strasbourg	A. Jimborean, Luis Mastrangelo , V. Loechner and P. Clauss. VMAD: an Advanced Dynamic Program Analysis & Instrumentation Framework . 2012
INRIA/University of Strasbourg	A. Jimborean, P. Clauss, B. Pradelle, Luis Mastrangelo and V. Loechner. Adapting the Polyhedral Model for Efficient Speculative Parallelization . 2012